

RESEARCH ARTICLE

A novel COVID-19 sentiment analysis in Turkish based on the combination of convolutional neural network and bidirectional long-short term memory on Twitter

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Abstract

The whole world has been experiencing the COVID-19 pandemic since December 2019. During the pandemic, a new life has been started by necessity where people have extensively used social media to express their feelings, and find information. *Twitter* was used as the source of what people have shared regarding the COVID-19 pandemic. Sentiment analysis deals with the extraction of the sentiment of a given text. Most of the related works deal with sentiment analysis in English, while studies for Turkish sentiment analysis lack in the research field. To this end, a novel sentiment analysis model based on the combination of *convolutional neural network* and *bidirectional long short-term memory* was proposed in this study. The proposed deep neural network model was trained on the constructed *Twitter* dataset, which consists of 15k Turkish tweets regarding the COVID-19 pandemic, to classify a given tweet into three sentiment classes, namely, (i) *positive*, (ii) *negative*, and (iii) *neutral*. A set of experiments were conducted for the evaluation of the proposed model. According to the experimental result, the proposed model obtained an accuracy as high as 97.895%, which outperformed the state-of-the-art baseline models for sentiment analysis of tweets in Turkish.

KEYWORDS

COVID-19, deep neural network, sentiment analysis, text classification, *Twitter*

1 | INTRODUCTION

The whole world has been experiencing the heavy effects of the coronavirus disease 2019 (COVID-19) pandemic since December 2019. According to the *World Health Organization (WHO)*, as of June 5, 2021, more than 172 million cases and more than 3.7 million deaths have been confirmed on a global scale.¹ When it comes to Turkey, there have been more than 5.2 million confirmed cases of COVID-19 with 48 million deaths.¹ *Twitter*,^{*} one of the most popular microblogging platforms with having 192 million daily active users as of the fourth quarter of 2020,² lets registered users post status messages, which are called “tweets.” A tweet can contain characters up to 280 characters and can be concerned with everything related to life, including but not limited to economics, politics, personal life, products, health, and entertainment. In addition to the public, companies use social media (e.g., *Twitter*) to promote products, brand names, and services.³ Additionally, many governments use *Twitter* to communicate with the public.⁴ Especially during the pandemic, people have used social media to express their feelings, find information, and also calm themselves⁵ which is unsurprising as it was reported that people feeling lonely were more inclined to use social media to cope with lacking social contact during the COVID-19 lockdown.⁶ According to a recent report,⁷ *Twitter* saw a record number of users, 186 million, during the pandemic. Therefore, the proposed study

intentionally used *Twitter* as the source of the data. This huge number is self-representative regarding the interpretation of the data posted on *Twitter* is highly time-consuming and labor-intensive. Sentiment analysis is one of the hottest research topics of computer science and is accepted as a subfield of natural language processing (NLP). Sentiment analysis is basically defined as the task of extracting emotions or opinions from a given text for a given topic. Sentiment analysis is an increasingly popular instrument for the analysis of social media.⁸ The underlying technology of sentiment analysis is a part of artificial intelligence (AI) as the essence of it is a text classification task.⁹ *Twitter* sentiment analysis is a subfield of sentiment analysis that deals with identifying sentiments in tweets that combines the power of sentiment analysis with the popularity of *Twitter*. Due to these advantages, *Twitter* sentiment analysis has been widely adopted in many different domains, including but not limited to politics,¹⁰ economics,¹¹ product sales,¹² medical,¹³ sports analytics,¹⁴ security informatics,¹⁵ and consumer satisfaction and brand analysis.¹⁶ *Twitter* sentiment analysis is considered a more challenging task than sentiment analysis on conventional text (e.g., review documents) due to (i) the short length of tweet messages, (ii) the frequent use of informal and irregular words, slang, and misspelled words (e.g., repeated characters), (iii) the usage of *Twitter*-specific keywords such as hashtags, and usernames, (iv) the usage of emojis, and emoticons, and (v) the rapid evolution of the language in *Twitter*.^{17,18} Most of the related works deal with sentiment analysis in English, while studies for Turkish sentiment analysis lack in the research field. To this end, a novel *Twitter* sentiment analysis model based on the combination of *convolutional neural network (CNN)* and *bidirectional long short-term memory (BiLSTM)* for Turkish was proposed with this study. The proposed deep neural network (DNN) model was trained and evaluated on the constructed dataset that consists of the collected Turkish tweets regarding the COVID-19 pandemic. The main contributions of the proposed study are listed as follows:

- A novel deep neural network, which is a combination of *CNN* and *BiLSTM*, was proposed for *Twitter* sentiment analysis in Turkish. The proposed model was finalized after the model optimization task, which was performed in an automated way thanks to the employed *Grid Search* technique.
- Unlike the related work, both emojis and emoticons were not cleared during the preprocessing since they were used within the proposed sentiment analysis model.
- The tweet annotation process was performed in an automated way instead of manual annotation, which is highly time-consuming and labor-intensive. Additionally, instead of labeling tweets with respect to whether they contain a predefined set of symbols or words (e.g., emojis, emoticons, keywords, and hashtags), the employed approach uses a lexicon-based tweet annotation.

The rest of the article is organized as follows: Section 2 briefly reviews the related work. Section 3 describes the material and method used for the proposed study. Section 4 describes the experimental result and discussion in the light of conducted experiments. Finally, Section 5 concludes the article with future directions.

2 | RELATED WORK

Coban et al.¹⁹ proposed an approach based on traditional machine learning (ML) algorithms for the Turkish sentiment analysis task in *Twitter*. They classified the tweets into two sentiment classes, namely, (i) *positive*, and (ii) *negative*. They queried the *Twitter* API through the predefined set of emoticons where containing the “: -)”, “: ;)”, “=)”, and “: D”, and “: -(”, “: (”, “= (”, and “: (” emoticons label the corresponding tweets as the *positive* and *negative*, respectively. Based on this assumption, they constructed a dataset that consisted of 20k Turkish tweets. Our study intentionally did not use this assumption since (i) a tweet may contain emoticons from both classes, and (ii) a tweet does not have to contain any emoticons. During the preprocessing, they remove emoticons from tweets. Unlike this study, we intentionally kept emoticons as well as emojis as they are sentimentally meaningful (as discussed in Section 3). They employed four traditional ML algorithms, namely, the (i) *support vector machine (SVM)*, (ii) *Naïve Bayes*, (iii) *Multinomial Naïve Bayes*, and (iv) *k-Nearest Neighbors (k-NN)*. According to their experimental result, the *Multinomial Naïve Bayes* classifier obtained the highest accuracy, an accuracy of 66.06% when the features of vector space were extracted through the employed *n*-gram method. Akgun et al.²⁰ proposed an approach based on lexicon and *n*-grams. The proposed method was evaluated on the constructed dataset that was labeled into three classes, namely, (i) *positive*, (ii) *negative*, and (iii) *neutral* through a given lexicon. According to the experimental result, the lexicon-based method slightly outperformed the *n*-gram based method by obtaining an *F1-score* of 70% compared to 69%. One drawback of this study is that, unlike our study, emotional expressions in tweets were removed during the preprocessing. Another drawback of this study is that they assigned the thresholds for the emotion classes in a not-formulaic way. Demirci²¹ proposed an approach for the emotion analysis of Turkish tweets and collected tweets for six emotions, namely, (i) *anger*, (ii) *disgust*, (iii) *fear*, (iv) *joy*, (v) *sadness*, and (vi) *surprise*. The constructed dataset contained 6000 tweets, 1000 tweets per emotion. The effects of various traditional ML algorithms, namely, (i) *Naïve Bayes*, (ii) *Complement Naïve Bayes*, (iii) *SVM*, and (iv) *k-NN*. According to the experimental result, the *SVM* outperformed the other algorithms by obtaining an accuracy of 69.92%. Ileri²² proposed a user-centric approach for Turkish emotion analysis in *Twitter*. The motivation behind this study was that connected users may be more likely to hold similar emotions. They considered the same six emotions that were considered by Demirci.²¹ The dataset that was used for this study was constructed by querying the *Twitter* API through the predefined Turkish keywords for the covered emotions as follows (English translations are given in the parenthesis): (i) “*sinir*” (“*anger*”), (ii) “*korku*”

("fear"), (iii) "mutluluk" ("happiness"), (iv) "üzüntü" ("sadness"), (v) "iğrenç" ("disgusting"), and (vi) "şaşkınlık" ("surprise"). Consequently, a total of 72,000 tweets were collected, 1200 tweets for each emotion. According to their experimental result, they reported that multi-class emotional domains, ideally with more than two emotions, cause lots of biases in class predictions unless the classes are greatly separable in terms of features. Unlike this study, we intentionally did not remove emotional expressions (e.g., emojis and emoticons) from tweets during preprocessing since they are sentimentally meaningful (as discussed in Section 3). Also, weighted edges were ignored in this study as all edge weights were assumed to equal to 1. Tocoglu et al.²³ proposed an approach based on DNNs for the emotion analysis of Turkish tweets. The six emotion classes they considered were the same as the emotion classes considered by Demirci.²¹ Since the dataset constructed by Demirci²¹ is not publicly available and large enough to be used in DNNs, Tocoglu et al.²³ constructed their own dataset that consists of 205,278 tweets. They labeled these tweets using a lexicon-based method. When it comes to the classification task, they proposed various models based on three different DNN architectures, namely, (i) *artificial neural network* (ANN), (ii) *CNN*, and (iii) *LSTM*. According to the experimental result, while the proposed *CNN* obtained the highest *accuracy*, an *accuracy* of 74%, the proposed *ANN* obtained the worst *accuracy*. The authors mentioned that they did not perform cross-validation due to the computational constraints of using a large dataset. Unlike this work, we performed *k-fold cross-validation*, which is a technique that ensures each observation in the dataset is used for both training and validation (more detail is provided in Section 4). Demirci et al.²⁴ proposed a sentiment analysis approach for Turkish using both traditional ML and deep learning techniques. They collected 3000 tweets regarding the coup attempt that happened on July 15, 2016. They covered two sentiment classes, namely, (i) *positive*, and (ii) *negative*. To classify the tweets, they employed three traditional ML techniques, namely, (i) *logistic regression*, (ii) *SVM*, and (iii) *random forest*, and a DNN model that consisted of six *Dense* and three *Dropout* layers. According to the experimental result, their DNN provided better performance than the traditional ML techniques by obtaining an *accuracy* of 81.86%. One drawback of this study is that, unlike our study, neither emojis nor emoticons were used within this study. Another drawback of this study is that their constructed dataset is too small to effectively train a DNN. Ucan et al.²⁵ proposed a Turkish emotion analysis based on four pre-trained language models, namely, (i) *BERT*, (ii) *DistilBERT*, (iii) *BERTurk*, and (iv) *DistilBERTurk*. They employed various traditional ML such as *SVM*, and *Naïve Bayes* and deep learning techniques such as *gated recurrent unit* (*GRU*), and *LSTM*. According to conducted experiments on various well-known datasets, the proposed model based on *BERTurk/BERT* obtained the best performance. One drawback of this study is the absence of the employment of the features that target the emotional contexts of tweets. In addition to the related work that was proposed for sentiment analysis in Turkish, there exist some studies that employ the combination of *CNN* and *LSTM* for sentiment analysis in various languages such as English, Vietnamese, and Arabic. Onan²⁶ proposed an architecture that combines TF-IDF weighted *Glove* word embedding with *CNN-LSTM* architecture for sentiment analysis on product reviews. The proposed architecture was trained and evaluated on a total of 6940 tweets written in English. The three sentiment classes that were covered in this study were, namely, (i) *positive*, (ii) *negative*, and (iii) *neutral*. This architecture obtained an *accuracy* of 93.85% on the test set of the employed dataset, which consisted of 692 tweets. Unlike this study, we have employed a hyperparameter optimization task to reveal the best-performing hyperparameters for the proposed DNN model. Vo et al.²⁷ proposed an approach based on the combination of *CNN* and *LSTM* for sentiment analysis in Vietnamese. To this end, they have collected comments/reviews from Vietnamese commercial web pages. The constructed corpus consisted of 17,500 reviews and was annotated by three human annotators. They covered three sentiment classes, namely, (i) *positive*, (ii) *negative*, and (iii) *neutral*. According to the experimental result, the proposed model outperformed *SVM*, *LSTM*, and *CNN* baseline models. One drawback of this study is that the proposed model does not employ text preprocessing techniques despite that it is necessary for Vietnamese. Ombabi et al.²⁸ proposed an approach based on the combination of *CNN-LSTM* and *SVM* for sentiment analysis in Arabic. They employed *fastText*,²⁹ an open-source library by *Facebook AI Research*, to construct the word embeddings. The feature maps were learned by the proposed *CNN-LSTM* model. Then, these features maps were yielded into the employed *SVM* to classify the given input into two sentiment classes, namely, (i) *positive*, and (ii) *negative*. According to the experimental result, the proposed model obtained an *accuracy* of 90.75% on a multi-domain corpus. One drawback of this study is that they empirically (without employing any well-known techniques) determined the values of hyperparameters. Instead of this approach, we employed a widely-used technique for this critical task (which is described in detail in Section 3.4). The comparison of the related work is given in Table 1.

3 | MATERIAL AND METHOD

In this section, we start by describing the details of the dataset construction process that was used by the proposed method. Then, the employed preprocessing process that was applied to the constructed dataset, and how this unlabeled dataset was annotated were described. Finally, the details of the proposed method are discussed.

3.1 | Dataset construction

The constructed dataset contains Turkish tweets related to the COVID-19 pandemic from *Twitter*. These tweets were fetched through the *Twitter Standard Search API*³⁰ that is provided by *Twitter*. To this end, an open-source Python library, namely, *tweepy*,³¹ was employed. The criteria that were

TABLE 1 The comparison of the related work in terms of (i) employed technique(s), (ii) covered emotions, and (iii) drawback(s)

Related work	Employed technique(s)	Covered emotions	Drawback(s)
Coban et al. ¹⁹	Traditional ML algorithms	(i) Positive, and (ii) negative	Labeling tweets through the predefined set of emoticons
Akgun et al. ²⁰	Lexicon-based techniques	(i) Positive, (ii) negative, and (iii) neutral	Removal of emotional expressions during preprocessing Assigning the thresholds for the emotion classes in a not-formulaic way
Demirci ²¹	Traditional ML algorithms	(i) Anger, (ii) disgust, (iii) fear, (iv) joy, (v) sadness, and (vi) surprise	Not employing DNNs alongside the traditional ML algorithms
Ileri ²²	User-centric techniques	(i) Anger, (ii) disgust, (iii) fear, (iv) joy, (v) sadness, and (vi) surprise	Removal of emotional expressions during preprocessing Ignorance of weighted edges as all edge weights were assumed to equal to 1
Tocoglu et al. ²³	Lexicon-based techniques and DNNs	(i) Anger, (ii) disgust, (iii) fear, (iv) joy, (v) sadness, and (vi) surprise	Not employing the cross-validation technique
Demirci et al. ²⁴	Traditional ML algorithms, and DNNs	(i) Positive, and (ii) negative	Removal of emotional expressions during preprocessing The constructed dataset is too small to effectively train a DNN
Ucan et al. ²⁵	Pre-trained language models, traditional ML algorithms, and DNNs	(i) Anger, (ii) disgust, (iii) fear, (iv) joy, (v) sadness, and (vi) surprise	Absence of the employment of the features that target the emotional contexts of tweets
Onan ²⁶	Pre-trained language models, and DNNs	(i) Positive, (ii) negative, and (iii) neutral	Absence of the hyperparameter optimization task
Vo et al. ²⁷	DNNs	(i) Positive, (ii) negative, and (iii) neutral	Absence of the employment of text preprocessing techniques
Ombabi et al. ²⁸	Pre-trained language models, traditional ML algorithms, and DNNs	(i) Positive, and (ii) negative	Hyperparameters of the proposed DNNs were empirically determined

set to filter tweets from the *Twitter Standard Search API* were as follows: (i) the tweets should be written in Turkish, (iii) the tweets should contain at least one of the following keywords: “*covid*”, “*covid-19*”, “*covid19*”, “*koronavirüs*”, and “*pandemi*”, and (iii) the tweets should not be “*retweets*”, which are the tweets that were broadcast by users to inform their network, to prevent duplications. Consequently, a total of 50k tweets were collected. The sequence length distribution of these tweets is presented in Figure 1.

3.2 | Preprocessing

Since the collected tweets are raw, a preprocessing process is key to preparing them to be ready to be yielded into the proposed DNN. The preprocessing process converts the data to a more meaningful form, which eventually makes the ML model better represent the data.³² As a natural result of this improvement, the accuracy of the ML model improves. The employed preprocessing process performs the following operations on the collected raw tweets:

- The capital letters of tweets were folded to lowercase.
- The hyperlinks, which are sentimentally meaningless, were cleared the tweets.
- The stop words, which are commonly used words in a natural language that do not convey polarity, were cleared from the tweets to reduce the noise in textual data.⁵ The stop words, that are specific to Turkish, were retrieved from an open-source Python library, namely, *stop-words*.³³ The keywords that were used to query the *Twitter Standard Search API* were also included in the list of stop words.
- A *Twitter* mention is a keyword that is a part of *Twitter*’s internal communication mechanism and contains a username of another *Twitter* user by preceding a “@” character. It is not a “must” for a tweet to contain mention(s) but a tweet may contain mentions as long as its character limit, 280

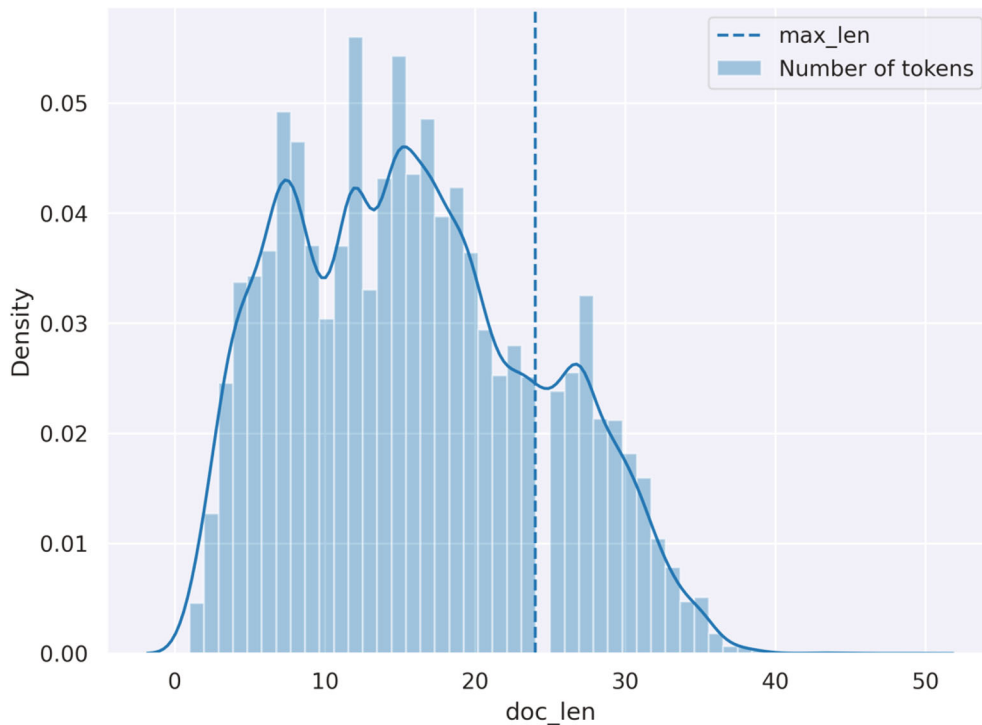


FIGURE 1 The sequence length distribution of the tweets in the constructed dataset

TABLE 2 The number of samples that each sentiment class of the final dataset consisted of

Sentiment class	Number of samples
Positive	3290
Negative	852
Neutral	45,802

characters, allows. Since Twitter mentions are sentimentally meaningless, they were cleared from the tweets. It is worth mentioning that the “@” character is not the only one that was cleared from tweets; but also the succeeding username since they may be sentimental despite they actually do not aim to be (e.g., “@happiness”).

- A Twitter hashtag is a keyword that tweets may contain to declare the topic of a tweet. A Twitter hashtag contains one word or combined words (without any space between these words) by preceding a “#” character. Since Twitter hashtags do not convey a sentimental polarity, they were cleared from tweets.
- The words, that contain a single character, were cleared from the tweets.
- The punctuation marks were cleared from the tweets after the extraction of emojis and emoticons.
- Unlike the related work, emojis and emoticons were intentionally not cleared from the tweets since they are sentimentally meaningful.^{34,35}

Six of the fetched 50k tweets were eliminated from the dataset since becoming empty Strings (“”) after the preprocessing process. Therefore, the final dataset consisted of 49,994 unique tweets. Any stemming approaches were not applied since stemming algorithms do not work well with tweets.²³ The number of samples that each sentiment class of the final dataset consisted of is listed in Table 2.

3.3 | Dataset annotation

Since the constructed dataset is not annotated, it is necessary to annotate it as the proposed model is based on a supervised ML technique. In literature, the datasets were commonly annotated based on either using a set of keywords^{21,36,37} or emoticons.³⁸⁻⁴⁰ Unlike these works, a lexicon-based approach was employed for generalization. When it comes to the state-of-the-art lexicons, *TextBlob*⁴¹ was intentionally not opted since it does not

utilize social media-specific symbols such as emojis, and emoticons while analyzing the sentiment of a given sentence. To this end, *VADER* (*Valence Aware Dictionary and sEntiment Reasoner*),⁴² which is a lexicon and rule-based sentiment analysis tool that is specifically attuned to sentiments expressed in social media, was employed. *VADER* basically reveals a polarity of a given sentence in the range of $[-1, +1]$ where -1 , and $+1$ mean extremely *negative* and extremely *positive* sentiments, respectively. Even though *VADER* was originally developed for sentiment analysis in English, we have employed it on Turkish tweets since (i) it also uses emotional expressions (e.g., emojis, and emoticons), slang, hashtags, and punctuation marks, which are all common in tweets, alongside words, and (ii) these features are language-independent. The labels were assigned based on the approach proposed by Bandi and Fellah⁴³ as follows: If the calculated polarity of a tweet is less than -0.4 , it is labeled as *negative*. Symmetrically, if the calculated polarity of a tweet is greater than $+0.4$, it is labeled as *positive*. The tweets whose polarity values are in the range from -0.4 to $+0.4$ are labeled as *neutral*. This approach is formulated in Formula 1 where T_i , P_i , and S_i indicate the given tweet T_i , the calculated polarity of T_i , and the assigned sentiment label of T_i , respectively.

$$S_i = \begin{cases} \text{negative,} & P_i < -0.4 \\ \text{neutral,} & -0.4 \leq P_i \leq +0.4 \\ \text{positive,} & P_i > +0.4 \end{cases} \quad (1)$$

3.4 | Proposed model

The proposed model is a novel DNN that is a combination of *CNN* and *BiLSTM* to classify a given tweet's sentiment into one of the three sentiment classes, namely, (i) *negative*, (ii) *neutral*, and (iii) *positive*. *CNNs* are basically composed of different types of layers. The proposed model was implemented with *Keras*,⁴⁴ a widely used deep learning API that was written in Python. *Keras* acts as a wrapper for various DNN backends. *TensorFlow*,⁴⁵ a widely used open-source ML platform that was developed by Google, was opted as the backend of the proposed model since it is the one recommended by the developer of *Keras*.⁴⁶ The proposed model consists of 13 layers as an overview of its architecture is presented in Figure 2. The model started with an *Embedding* layer, which creates vector space for the given input (tweets). Then, a 1-dimensional *Convolutional layer* (*Conv1D*) was employed to perform convolution operations on the output of the *Embedding* layer. While n , len_i , and m denote the number of input sequences, the number of tokens of the input sequence i , and the mean of the lengths of the input sequences, the calculation of the

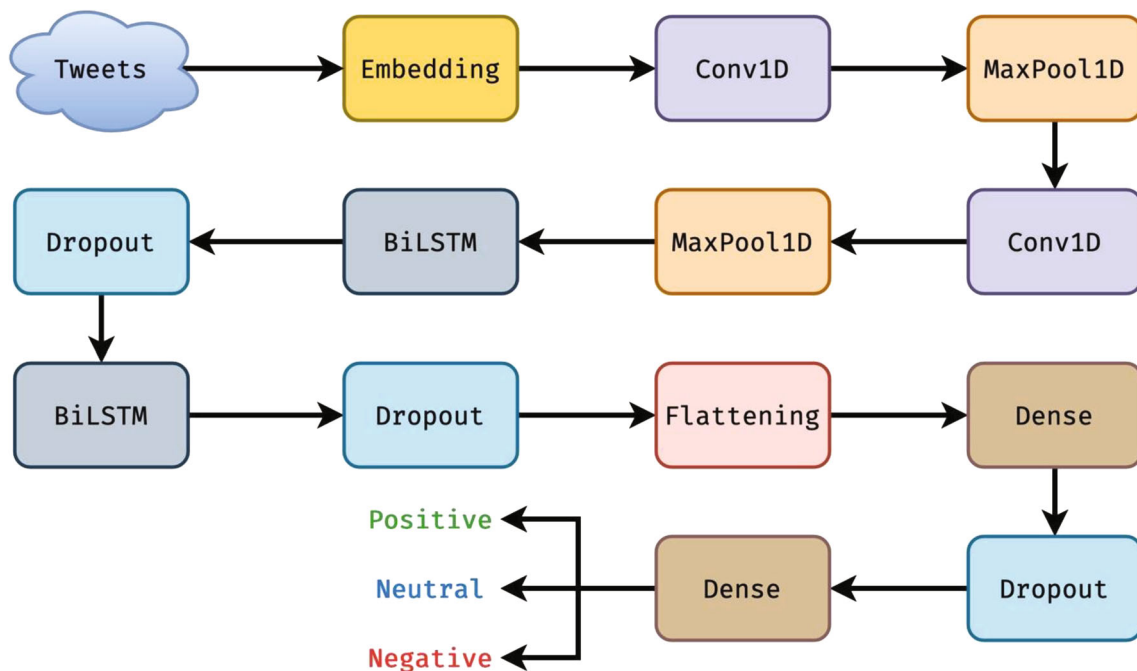


FIGURE 2 An overview of the architecture of the proposed DNN model, which is a combination of *CNN* and *BiLSTM*

length of the input sequences of the *Embedding* layer (denoted by max_len) was formulated in Formula 2. The max_len was calculated as 24 for the constructed dataset.

$$m = \frac{\sum_{i=1}^n len_i}{n}$$

$$max_len = m + \sqrt{\frac{\sum_{i=1}^n (len_i - m)^2}{n}} \quad (2)$$

Following the *Conv1D* layer, a 1-dimensional *Max Pooling* (*MaxPool1D*) layer was employed to reduce the spatial size of the representation that helps to reduce the number of parameters and computation in the network.⁴⁷ *Recurrent neural network* (*RNN*) is a type of feedforward ANN, which is capable of handling variable-length sequence inputs.⁴⁸ *LSTM*⁴⁸ is a special type of *RNN* that contains gates to maintain memory for long periods.⁴⁹ Therefore, *LSTM* is capable of remembering the past data and producing output with respect to the past and current data, which are key necessities for our case. The main advantages of *LSTM* over *RNN* are as follows: (i) *LSTM* has the ability to solve the general problem of gradient descent,⁵⁰ and (ii) it contains a long-term memory which is a key necessity for sequence processing. *Bidirectional LSTM* (*BiLSTM*) is a further development of *LSTM* that has the ability to access both preceding and succeeding contexts. Therefore, *BiLSTM* can solve sequential modeling better than *LSTM*.⁵¹ Due to the aforementioned advantages of *BiLSTM*, it was intentionally employed in the proposed model instead of *RNN* and *LSTM*. After these typical layers of *CNN*, a *BiLSTM* layer was employed. Then, a *Dropout*⁵² layer was employed to randomly drop a defined ratio of neurons from the neural network. Therefore, *Dropout* is a widely used technique to prevent the “overfitting” problem, which is one of the major challenges of DNNs because of increased depth and complexity.^{53,54} This pattern (sequentially added *BiLSTM* and *Dropout* layers) was employed one more time. Then, a *Flattening* layer was employed to reshape the input (a matrix) into a vector. Following the *Flattening* layer, two *Dense* layers, which are fully connected neural network components, were employed for classification purposes. Between these two *Dense* layers, a *Dropout* layer was employed. The *Softmax* was employed as the activation function of the last *Dense* layer, which is responsible for the classification of the given tweet into one of the three sentiment classes. Except for the last *Dense* layer, the *ReLU* was employed as the activation function for all *Conv1D* layers and the other *Dense* layer to introduce non-linearity to the proposed DNN. The *Adaptive Moment Estimation* (*Adam*)⁵⁵ algorithm, which is an extension to *Stochastic Gradient Descent* (*SGD*), was employed as the optimization algorithm of the proposed neural network to update the neural network weights more efficiently by computing adaptive learning rates for each network parameter from estimates of the first and second moments of the gradient.⁵⁶ A loss function in a neural network is responsible for estimating the loss in each iteration (*a.k.a.* epoch) in order to reduce it on the next iteration. The *Categorical Cross-entropy* function was employed as the loss function since the task of the proposed neural network is a multi-class classification problem.

Hyper-parameters of a neural network are the parameters of a DNN model that affect the learning process and are set empirically. Therefore, the *Grid Search* technique, which is an extensive way of hyper-parameter optimization since all possible combinations of hyper-parameter values were literally experimented with,⁵⁷ was employed to reveal the best value for each hyper-parameter. To this end, each hyper-parameter has experimented with various values as they are listed in Table 3. This technique was realized thanks to the *scikit-learn*⁵⁸ library as it provides an easy-to-use integration with *Keras* to programmatically evaluate the given values for the hyper-parameters and then reveal the best combination of them that produces the highest accuracy, which was regarded as the success metric of the proposed model. As a result of this process, the employed hyper-parameters were determined. The layers of the proposed model including the corresponding hyper-parameters are listed in Table 4.

TABLE 3 The evaluated values for the hyper-parameters of the proposed model

Hyper-parameter	Evaluated values
Number of filters of the <i>Conv1D</i> layers	32, 64, 128
Kernel size of the <i>Conv1D</i> layers	3, 5, 7
Number of units of the <i>Dense</i> layers	32, 64, 128, 256
Optimization algorithm	<i>Adam</i> , <i>SGD</i> , <i>RMSprop</i> , <i>Adamax</i>
Learning rate of the optimization algorithm	$1 \times e^{-4}$, $1 \times e^{-3}$, $1 \times e^{-2}$
Size of vector space of the <i>Embedding</i> layer	32, 64, 128, 256
Activation function (except for the last <i>Dense</i> layer)	<i>ReLU</i> , <i>tanh</i> , <i>sigmoid</i> , <i>softmax</i>
Batch size	32, 64, 128
Dropout rate	0.25, 0.3, 0.4, 0.5, 0.6, 0.7

TABLE 4 The layers of the proposed model, including the employed hyper-parameters

Layer #	Layer type	Hyper-parameters with their values
1	<i>Embedding</i>	• Vocabulary size: 91,012 • Dimension of embedding: 64 • Length of input sequences: 24
2	<i>Conv1D</i>	• Number of filters: 64 • Kernel size: 3 • Activation function: <i>ReLU</i>
3	<i>MaxPool1D</i>	• Pool size: 2
4	<i>Conv1D</i>	• Number of filters: 64 • Kernel size: 3 • Activation function: <i>ReLU</i>
5	<i>MaxPool1D</i>	• Pool size: 2
6	<i>BiLSTM</i>	• Number of units: 10
7	<i>Dropout</i>	• Dropout rate: 0.6
8	<i>BiLSTM</i>	• Number of units: 10
9	<i>Dropout</i>	• Dropout rate: 0.6
10	<i>Flattening</i>	—
11	<i>Dense</i>	• Number of units: 32 • Activation function: <i>ReLU</i>
12	<i>Dropout</i>	• Dropout rate: 0.6
13	<i>Dense</i>	• Number of units: 3 • Activation function: <i>Softmax</i>
<i>Adam</i>		• Learning rate: $1 \times e^{-3}$

4 | EXPERIMENTAL RESULT AND DISCUSSION

The whole dataset was split as follows: 70% and 30% of the whole dataset were used as training and validation sets (which consisted of 34,960 samples), and test sets (which consisted of 14,984 samples), respectively, due to being the commonly used ratio in data mining.⁵⁹ When it comes to the split of the training and validation sets, the *stratified k-Fold cross-validator*, which ensures that generated test sets contain the same (or as close as possible) distribution of target classes, was employed to create folds. The *stratified k-Fold cross-validator* was intentionally opted instead of the *k-Fold cross-validator* since the distribution of the target classes in the constructed dataset was imbalanced.⁶⁰ The *stratified k-Fold cross-validator* was implemented thanks to the *scikit-learn* library. The number of folds (*k*) was set to 10 which means that the training and validation sets were divided into ten equal parts where nine parts (90% of the training and validation sets, which equals 31,464) were used for training, and the remaining one part (10% of the whole set, which equals 3496) was used for validation. The training of the proposed DNN was started with the *Early Stopping*⁶¹ callback, which helps to prevent overfitting⁶² by stopping the training when the monitored criterion (*a.k.a.* monitor) has not been getting better for the predefined number of epochs (*a.k.a.* patience). The *monitor*, and *patience* of the proposed model were defined as the calculated loss of the validation set (*a.k.a.* validation loss) and 2 epochs. The batch size of the training set, which defines the number of training samples used to estimate the gradient direction before the model's internal parameters are updated, was set to 64 as a result of the employed *Grid Search*. The training had continued for 50 epochs as the plots of the calculated accuracy values for the training and validation sets over the epochs were given in Figure 3. As it can be seen from this figure, it is safe to say that the proposed model does not overfit.

After the training of the proposed model was completed, it was evaluated on the test set whose samples were used for neither training nor validation purposes. Since the proposed model handles a classification problem, its efficiency was measured through a de-facto standard technique, namely, *confusion matrix*. This technique defines several evaluation metrics including but not limited to *accuracy*, *precision*, *F1-score*, and *recall*. While *T*, *F*, *P*, and *N* denote correctly classified samples, misclassified samples, samples labeled as *positive*, and samples not labeled as *positive*, respectively, the aforementioned evaluation metrics were formulized in Formula 3.

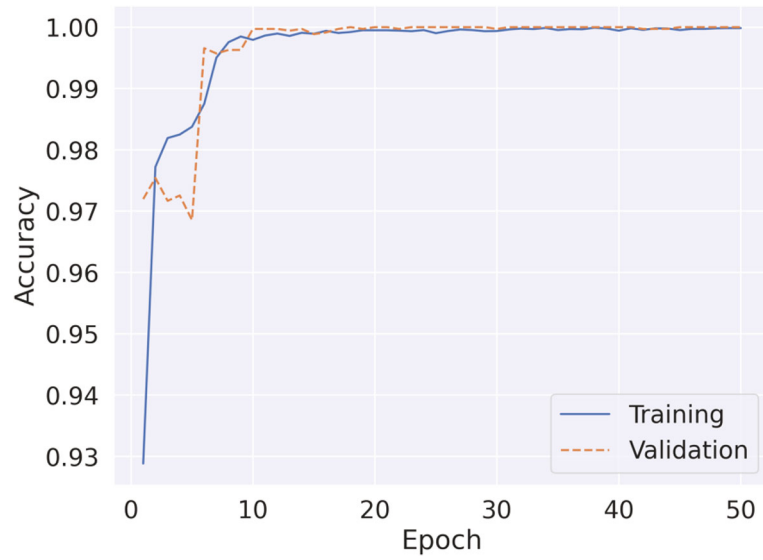


FIGURE 3 The plots of the calculated *accuracy* values for the training and validation sets over the epochs

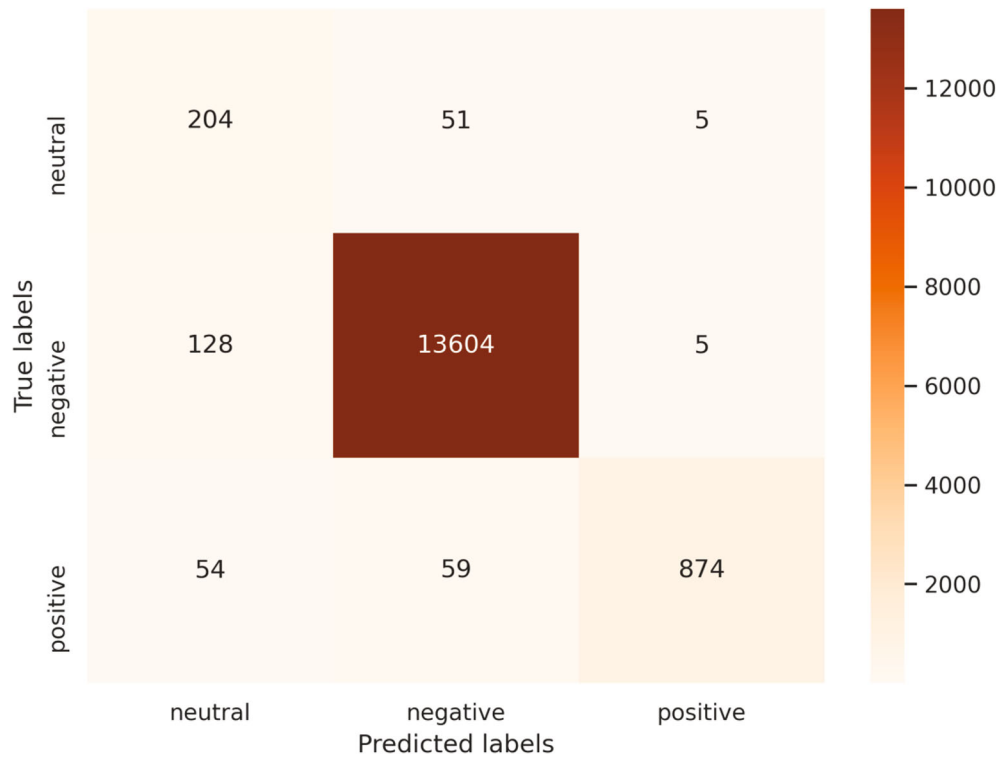


FIGURE 4 The confusion matrix of the evaluation of the proposed model on the test set

$$\text{Accuracy} = \frac{TP + TN}{P + N}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{F1-score} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

$$\text{Recall} = \frac{TP}{P}$$

TABLE 6 The comparison of the proposed model with the baseline models in terms of their efficiencies of classifying the given tweets

Related work	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
COVIDSenti_CNN ⁵	89.442	93.015	98.073	89.442
COVIDSenti_LSTM ⁵	97.691	97.805	98.015	97.691
Proposed model	97.895	98.116	98.372	97.985

5 | CONCLUSION

Social networks have been an essential part of people's daily lives, especially during the COVID-19 pandemic as many countries have declared full or partial lockdown for long periods. Given a vast number of posts that have been shared daily on social networks, it is highly time-consuming and labor-intensive to interpret their content by hand. Sentiment analysis based on text mining has demonstrated its efficiency in many domains thanks to the advances in AI and NLP. In this study, a novel Twitter sentiment analysis model for Turkish, which is a DNN based on a combination of CNN and BiLSTM, was proposed. The proposed model can be easily adopted to various AI-powered tasks, including but not limited to (i) public opinion pooling, (ii) brand/product confidence surveying, (iii) understanding election tendencies, and customer analysis, and (iv) marketing optimization. To train the proposed DNN, a dataset, that consisted of 15k unique (RTs were excluded) Turkish tweets, was constructed. The proposed model was both trained and evaluated on the constructed dataset after splitting the whole dataset into training, validation, and test sets. According to the experimental result, the proposed model obtained an accuracy as high as 97.895% which outperformed the state-of-the-art baselines. The experimental result proves that the combination of CNN and BiLSTM provides a promising architecture for a challenging text analysis task such as Twitter sentiment analysis. As future work, the word embedding techniques can be employed before yielding the input data (the preprocessed tweets) into the proposed DNN. Also, an attention mechanism can be integrated into the proposed model in order to try to further improve the classification accuracy of the proposed model.

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CONFLICT OF INTEREST

The author declares that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ENDNOTES

^{*} <https://twitter.com>

[†] Turkish translation of "coronavirus".

[‡] Turkish translation of "pandemic".

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